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SCHOOL OF GRADUATE STUDIES

FACULTY OF COMPUTING

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Annotation Guidline

An Explainable Multi-Task Transformer Framework for Sentiment, Emotion, and Stance Detection in Amharic-English Code-Switched Social Media Text

1. Purpose and Scope

This guideline explains how to annotate Amharic-English code-switched social media comments collected from Facebook, YouTube, and Telegram for the research project:

An Explainable Multi-Task Transformer Framework for Sentiment, Emotion, and Stance Detection in Amharic-English Code-Switched Social Media Text.

Annotate only comments that combine **Amharic (Ge'ez script) and English within the same text**. Pure Amharic, pure English, and Romanized-Amharic-only comments are outside the study scope.

Each comment must be annotated for:

- Sentiment (one label)
- Emotion (one or more labels)
- Stance (one label toward an identified target)
- Supporting textual evidence for each task

2. General Annotation Rules

- Read the entire comment before assigning labels.
- Consider the combined meaning of Amharic and English.
- Annotate the overall meaning, not individual words in isolation.
- Interpret sarcasm and indirect expressions according to their intended meaning.
- Do not assume that sentiment, emotion, and stance are the same.
- Select the exact words or phrases that support each assigned label.
- Do not guess when the meaning or target is unclear. Flag the comment for review.

3. Task 1: Sentiment Annotation

Assign **exactly ONE** sentiment label to each comment.

Label	Definition
Positive	Expresses praise, approval, satisfaction, happiness, or support.
Negative	Expresses criticism, dissatisfaction, dislike, or disapproval.
Neutral	Does not express a clear positive or negative opinion.

Example:
Text: ይህ policy በጣም ጥሩ ነው.

Label: Positive

Supporting evidence: Highlight the words or phrase that express the positive or negative opinion. For Neutral, identify the relevant informational or non-opinion context.

4. Task 2: Emotion Annotation

Select **ONE OR MORE** emotion labels when applicable. Use Neutral only when none of the other emotions is expressed.

Label	**Definition**
Joy	Happiness, excitement, or pleasure.
Love	Affection, admiration, or attachment.
Anger	Frustration, irritation, or strong anger.
Sadness	Sorrow, grief, or disappointment.
Fear	Worry, anxiety, or a sense of danger.
Surprise	Shock or an unexpected reaction.
Disgust	Strong dislike or rejection.
Neutral	No identifiable emotional expression.

Example:
Text: በጣም happy ነኝ about this achievement!

Labels: Joy

Supporting evidence: Highlight the words or phrases that express each selected emotion. If Neutral is selected, identify the relevant text and confirm that no other listed emotion is expressed.

5. Task 3: Stance Annotation

Stance identifies the author's position toward a **specific target**, such as a person, government policy, political issue, organization, public event, or social topic.

For each comment, identify the target and assign exactly ONE stance label.

Label	Definition
FAVOR	The author supports, agrees with, or expresses approval toward the target.
AGAINST	The author opposes, disagrees with, or criticizes the target.
NONE	The comment expresses neither clear support nor opposition toward the target.

Example:

Text: I support this policy ምክንያቱም it helps our community.

- Stance target: This policy
- Stance: FAVOR
- Supporting evidence: Highlight the phrase expressing support for the policy.

Important: Stance must be determined toward the identified target. Do not automatically assign AGAINST because a comment has negative sentiment, or FAVOR because it has positive sentiment.

If no clear target can be identified, flag the comment for review instead of inventing a target.

6. Supporting Textual Evidence

For every comment, provide supporting textual evidence for all three tasks:

1. **Sentiment evidence:** Highlight the exact words or phrases supporting the sentiment label.
2. **Emotion evidence:** Highlight the words or phrases expressing the selected emotion(s).
3. **Stance evidence:** Highlight the words or phrases showing support, opposition, or the absence of a clear position toward the identified target.

Evidence must come directly from the original comment. Do not rewrite, translate, or invent evidence.

When a Neutral or NONE label has no explicit emotional or opinion cue, identify the relevant text context and do not invent supporting expressions.

7. Special Annotation Rules

Mixed sentiment: If a comment contains both positive and negative opinions, select the dominant or concluding opinion.

Sarcasm and irony: Annotate the intended meaning rather than the literal words.

Multiple emotions: Select all applicable emotions. Do not select Neutral together with another emotion.

Multiple targets: Identify the main target being discussed and annotate the author's stance toward that target. Flag cases where the target cannot be determined.

Unclear comments: Do not guess. Add a short comment explaining the uncertainty and flag the example for review.

Non-annotatable content: Skip comments that contain only emojis, symbols, bare links, usernames, or meaningless text without sufficient linguistic content.

8. Annotation Procedure

Follow this order for every comment:

1. Read and understand the complete Amharic-English code-switched text.
2. Assign one sentiment label.
3. Select one or more emotion labels.
4. Identify the stance target.
5. Assign one stance label: FAVOR, AGAINST, or NONE.
6. Highlight supporting textual evidence for sentiment, emotion, and stance.
7. Review all annotations before submitting.

9. Quality Control

- Three annotators will independently annotate the selected comments.
- Annotators should discuss the guideline and complete an initial calibration round before full annotation.
- Unclear cases and disagreements should be flagged for review and adjudication.
- Annotators should apply the same definitions and rules consistently.

10. Final Annotation Checklist

Before submitting each comment, confirm:

- The text contains both Amharic and English.
- Exactly one sentiment label is selected.
- One or more emotion labels are selected, with Neutral used only when appropriate.
- A stance target is identified.
- Exactly one stance label is selected.

- Supporting evidence is provided for sentiment, emotion, and stance.
- Evidence comes directly from the original comment.
- Unclear cases are flagged rather than guessed.

Thank you for contributing to this research.